

PhilFreela: Identity Verification (eKYC)

Tools, pretrained AI models, and services behind the ID check, face match, and live face scan

Prepared October 2, 2026. Versions are the ones installed and tested in the project on that date.

1. In short

eKYC (electronic know-your-customer) means confirming who a person is online. On PhilFreela, users verify with a government ID and a live face scan. Pretrained AI models check the photos, match the face on the ID with the live face, and detect a real head turn. **Nothing was trained or fine-tuned by us.** An admin makes every approval.

The AI cannot tell whether an ID card is genuine or fake. It checks that the ID photo, the live face, and the account fit together, not that the card was issued by the government. That last check is done by the admin, using the official government check where one exists (section 4).

2. How the verification works

The browser, the AI service, and the database each do a different job.

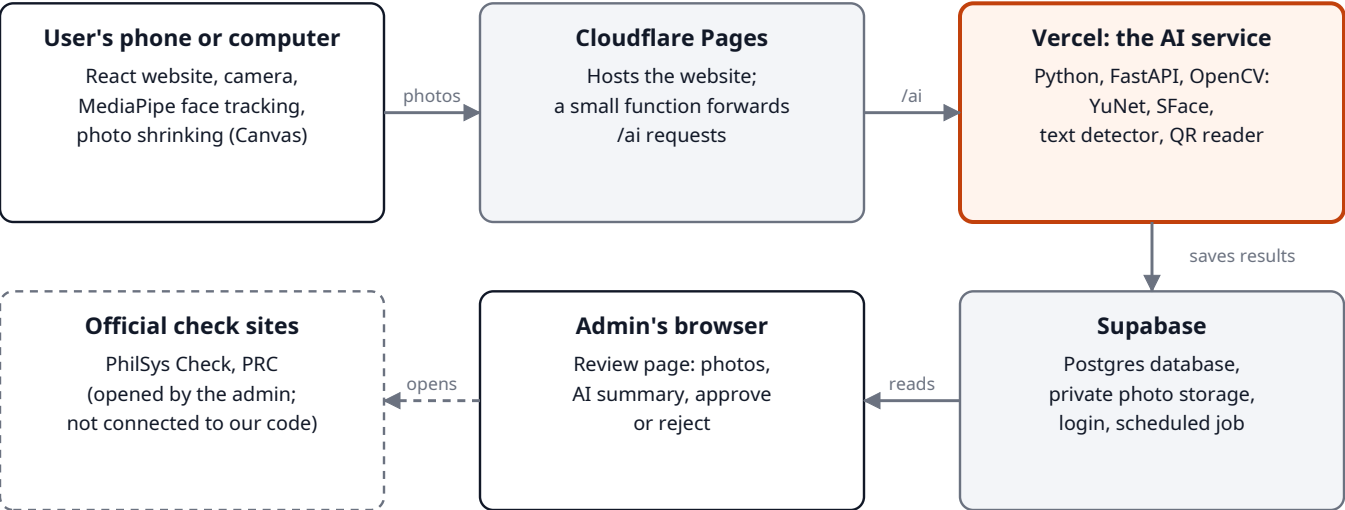


Figure 1. Photos travel from the user's browser through Cloudflare to the AI service on Vercel, which saves the results in Supabase. The admin reads them from Supabase and may open the official check sites by hand.

Step	What happens	Tools and AI used
1	The user picks an ID type (PhilSys, Driver's License, UMID, PRC or Passport). On a computer with a webcam a QR code is shown, so the same steps can be done on a phone.	React form; qrcode.react. The QR link works once and expires after 10 minutes.
2	Front of the ID. Instant check: a face is found and big enough, the photo is sharp, and it looks like an ID (enough printed text). The user cannot continue until it passes.	YuNet (face), Laplacian variance (sharpness), PP-OCrv3 text detector (lines of text)
3	Back of the ID (not for passports). Instant check: sharp, not the same photo as the front, enough printed text.	Same tools
4	Live face scan. Look straight, then turn the head all the way to each side (45° or more). Photos are taken automatically. If the face is lost, the scan restarts.	MediaPipe Face Landmarker (head angle in degrees); the browser camera
5	Review and consent. Photos are shrunk in the browser; the user agrees to the Data Privacy Act notice.	Canvas (browser), React

Step	What happens	Tools and AI used
6	The AI service re-runs every check (the browser is not trusted), then: matches the ID face with the live face; checks the same person appears during the whole scan; looks for the same face on another account; for PhilSys IDs, reads the QR code and compares its name with the profile name.	OpenCV with YuNet and SFace; OpenCV QR detector
7	Photos are saved as clean JPEGs (location data removed) in a private bucket, and the results in the database. The reply to the user only says the request was received.	Pillow, Supabase Storage, Postgres
8	An admin reviews the photos and an AI summary (Looks good, Check carefully, or Likely reject), then approves or rejects with a reason. Links to the official checks appear for PhilSys and PRC IDs.	React admin page; photo links that expire after 5 minutes
9	If no admin acts within 3 hours and the AI marked the request Likely reject, it is rejected automatically so the user can try again. The AI never approves.	Postgres function and pg_cron (runs every 15 minutes)
10	Approved users get a Verified check next to their name. Freelancers can post services only when verified.	Postgres Row Level Security rules

3. Tools used

3.1 Website (runs in the user's browser)

Tool	Version	Used for
React	19.3.0	The verification form: ID type, photo steps, face scan screen, review
Vite	8.2.2	Builds the website; development server
Bootstrap and Bootstrap Icons	5.3.3 and 1.11.3	Page layout and icons
MediaPipe Tasks Vision (Face Landmarker)	1.0.1	Tracks the face live in the browser (478 points) and works out how far the head is turned, in degrees
qrcode.react	4.2.0	QR code that opens the verification on a phone
React Router	7.18.3	Pages: Verify Identity, the phone page, and the admin Verifications page
Supabase JS Client	2.116.0	Login session; reads the user's verification status
Browser features	-	Camera (getUserMedia), Canvas (shrinking photos, taking scan snapshots), file picker (camera or gallery on phones)
@vitejs/plugin-basic-ssl	2.3.0	HTTPS while developing, because phones only allow the camera on HTTPS

3.2 AI service (Python, hosted on Vercel)

Tool	Version	Used for
Python	3.10.11	Language of the AI service
FastAPI	0.141.1	Web API: instant photo checks, the submit endpoint, and the phone QR link
Uvicorn	0.54.0	Runs the API on a laptop (Vercel runs it by itself)
Pydantic and python-multipart	2.13.5 and 0.0.32	Validate request data; receive photo uploads
OpenCV (opencv-python-headless)	5.0.0.93	Runs the three ONNX models, reads QR codes, measures sharpness
NumPy	2.2.6	Image arrays; distances between faces
Pillow	12.3.0	Opens photos, fixes rotation, re-saves clean JPEGs

Tool	Version	Used for
supabase-py and python-dotenv	2.31.0 and 1.2.3	Saves photos and results (secret service key kept on the server only); loads settings

3.3 Pretrained AI models

Model	Made by	Runs in	Size	License	Used for
MediaPipe Face Landmarker	Google	Browser	3.7 MB	Apache 2.0*	Live face tracking; the head-turn angle (liveness)
YuNet (2023mar)	Shiqi Yu's group, via OpenCV Zoo	AI service	0.2 MB	MIT	Finds faces and 5 points (eyes, nose, mouth corners) in ID photos and scan frames
SFace (2021dec)	Yaoyao Zhong et al., via OpenCV Zoo	AI service	37 MB	Apache 2.0	Turns a face into 128 numbers; faces are compared by cosine distance
PP-OCRRv3 text detector (English)	PaddleOCR, via OpenCV Zoo	AI service	2.3 MB	Apache 2.0	Finds lines of printed text (it does not read them), to tell an ID from other photos
OpenCV QR detector	OpenCV	AI service	built in	Apache 2.0	Reads the QR code on PhilSys IDs

Nothing was trained or fine-tuned by us. Every model is used as published. A face "embedding" is a list of numbers that describes a face; two photos of the same person give similar lists, so a small distance between the lists means "same person". Also used, with no model: Laplacian variance measures sharpness (sharp photos have big jumps in brightness between neighboring pixels), and a 32 by 32 pixel comparison spots a back photo that is the same as the front. *The MediaPipe library is Apache 2.0; its model cards are linked in the references.

3.4 Data, hosting, and automation

Tool	Plan	Used for
Supabase	Free; Postgres 17	Database (identity_verifications, verification_links); Auth (login tokens); Storage (private bucket for the photos); Row Level Security; database functions
pg_cron (Supabase extension)	–	Runs the 3-hour auto-reject check every 15 minutes
Cloudflare Pages	Free	Hosts the website; a Pages Function forwards /ai requests to the AI service
Vercel	Hobby (free), Tokyo region	Hosts the AI service; redeploys by itself when code is pushed
GitHub	–	Stores the code; a push updates Cloudflare and Vercel
PhilSys Check and PRC Verification	Official, free	Websites the admin opens by hand to check PhilSys and PRC IDs (not connected to our code)

3.5 Development and testing

Tool	Version	Used for
Visual Studio Code, Git, GitHub	Git 2.55.0	Code editor and version control
Node.js and npm	24.20.0 and 11.19.0	Runs the website tooling
Oxlint	1.82.0	Checks code quality
Playwright	1.63.0	Browser tests with a recorded camera video and simulated phones
Supabase Dashboard and Supabase MCP	–	Database management

Tool	Version	Used for
Claude Code (Anthropic)	–	AI coding assistant used during development

4. What the AI can and cannot tell, by ID type

Every ID type gets the same AI checks. They show that the photos are usable and that the face, the live scan, and the account fit together. They do not show that the card is real.

- A face is on the ID and is big enough, and the photo is sharp
- The photo looks like an ID (enough printed text), and the back is not the front again
- The face on the ID matches the live face
- A live person turned their head fully to both sides, and the same face appears during the scan
- The same face is not already on another account

ID type	Extra AI check	Can the AI tell real from fake?	What confirms the ID is real
PhilSys ID and ePhilID	Reads the QR code and compares its name with the profile name	No. The QR is digitally signed by the PSA (EdDSA). The PSA makes the public key needed to check the signature available to approved organizations (relying parties); we found no public download.	The admin scans the QR on the PSA's PhilSys Check site. A link is in the admin's AI card.
PRC ID	None	No.	The admin checks the license number and birthday on the PRC's free verification site. A link is in the AI card.
Driver's License (LTO)	None	No.	The admin judges the photos. We found no public tool for checking a license; the card's QR is described as something enforcers scan.
UMID	None	No.	The admin judges the photos. We found no public online tool; the issuing agency (SSS or GSIS) can confirm.
Passport	None (no back photo)	No. The ePassport chip holds a digital signature, but a website cannot read the chip from a photo.	The admin judges the photos. We found no public check.

Why the AI cannot do it. Telling a real ID from a good fake needs one of three things: the issuer's records or signing keys (the PSA gives these to approved organizations); special hardware such as a chip reader or UV light (not possible from a photo); or an AI trained on many real and fake IDs (this needs real ID photos, which are private data, and training, which this project does not do: pretrained models only).

What the AI suggests to the admin. Looks good: strong face match, live scan passed, nothing flagged. **Check carefully:** weak face match, the QR name does not match the profile, or the same face is on another account. **Likely reject:** no face match, or the live scan failed. The AI acts on its own in one case only: a Likely reject request that no admin has reviewed for 3 hours is rejected, so the user can try again. It never approves.

Sources for the table. Official sites: [PhilSys Check](#) and [PRC verification](#). News and commentary pages (not statements from the agencies): [PNA on the PhilSys QR signature](#), [AutoIndustriya on the LTO license](#), [Respicio on UMID](#).

5. The plan in the paper and what was built

The capstone paper names OCR, FaceNet, and liveness detection for eKYC. This table shows how each part was built.

Plan in the paper	What was built	Note
OCR reads the name and ID details	A text detector finds lines of printed text, to tell an ID from other photos. For PhilSys IDs the QR code is read and its name is compared with the profile name.	Reading the printed text of the other ID types (full OCR) is not built, so names are compared only for PhilSys IDs.
FaceNet compares the ID face with the selfie	SFace, run by OpenCV	Same idea: a face becomes a list of numbers and two lists are compared by distance. SFace is small enough for free hosting.
Liveness detection	A live head-turn challenge: look straight, then turn fully to each side, measured by MediaPipe	Stops printed photos and photos shown on a screen, which cannot turn. A recorded video could still fool it (section 9).
Verification badge	A blue check beside names; freelancers must be verified to post services	Clients may post jobs without verifying and show a Not verified label.
Freelancers verify during registration	Verification is a separate step after sign-up (Settings, then Verify now). Both freelancers and clients can do it.	Clients were an open item in the paper. Unverified users can still sign in; freelancers cannot post services until verified.

6. Security and privacy

Protection	How it works
Private storage	ID and face photos are in a private Supabase Storage bucket. Only admins can open them, through links that expire after 5 minutes.
Checks on the server	The AI service re-runs every check and confirms the user's login token with Supabase. The browser's own result is never trusted.
Write access	Only the AI service, using a secret service key that stays on the server and never reaches the website, can create verification records. Users cannot create or edit them. Admins can only change the status, the note, and the review details.
Clean photos	Photos are re-saved as new JPEGs (largest side 1280 pixels), so hidden data such as the GPS location is not copied.
Upload limits	JPEG, PNG, or WebP only (checked by content, not by file name), up to 5 MB each. One pending or approved verification per user.
Phone link	The QR link works once, expires after 10 minutes, and is replaced when a new one is made.
Consent	Users tick a consent box that cites the Data Privacy Act of 2012 (RA 10173) and mentions the duplicate-face check.
Face numbers	The 128-number description of the face is stored with the verification only to spot the same face on another account. It is removed with the verification record.
Accountability	Every decision records the admin who made it, or marks it as decided by the AI.

7. How the settings were chosen and tested

Check	Setting	Where the number comes from
Face match	Cosine distance 0.637 or lower	The cut-off published with the SFace model (cosine similarity 0.363 or higher). On about 60 public sample photos (149 same-person pairs and 1,596 different-person pairs, labeled with ArcFace as a reference), no different-person pair was accepted and 7 of the 149 same-person pairs (4.7%) were rejected.
Strong match and duplicate face	Cosine distance 0.50 or lower	Stricter than the match cut-off, so look-alikes are not flagged. In the same sample, different people were never closer than 0.64.
Live head turn	Straight: within 10°. Full turn: 45° or more, held for 4 frames. Halfway photo: 15° or more.	Sample photos of people facing the camera measured up to about 8°. A 45° turn is roughly looking toward the shoulder. Tested with a recorded video, not yet with many real users.
Face size on the ID	At least 120 pixels wide	In an early test, an ID photo whose face was about 100 pixels wide did not match.
Sharpness	Laplacian variance: face on the ID 25 or more; back photo 15 or more	Sharp sample photos scored 175 or more (faces) and 83 or more (backs); blurred copies scored 4 to 60 and 2 to 24.
Looks like an ID	Front: 8 or more lines of printed text. Back: 5 or more.	Sample ID cards (made-up names) had 14 to 15 lines on the front and 11 to 12 on the back; 63 selfies and portraits had 0 to 6.
Auto-reject wait	3 hours; the job runs every 15 minutes	A design choice: long enough for an admin to respond, short enough for the user to try again the same day.

Testing. During development, automated tests used made-up ID cards, public sample face photos, a recorded video as the camera, and simulated phones: 33 checks of the AI service's rules and 15 checks of the QR, duplicate-face, and auto-reject features (both sets also passed against the live Vercel service), plus tests of the verification form and the admin review page in a real browser. These show the checks behave as designed. They are not a study of real-world accuracy, which has not been done.

8. Tools tried and replaced

Earlier tool lists that show DeepFace, TensorFlow, RetinaFace, or ArcFace are out of date for identity verification. They were replaced during development for the reasons below.

First choice	Replaced by	Why
DeepFace with ArcFace and TensorFlow (RetinaFace as the backup face finder)	OpenCV with YuNet and SFace	TensorFlow made the AI service too big and slow for free hosting. On the development laptop it took roughly 20 seconds to load (now about 3), and RetinaFace took about 10 seconds per photo (YuNet: well under a second).
Hugging Face Spaces (Docker)	Vercel	Hugging Face's free plan no longer allows Docker Spaces.
ngrok tunnel to the developer's laptop	Vercel	It worked only while the laptop was on, and some Wi-Fi networks could not reach ngrok links.
Head turn estimated from the nose position	Head angle in degrees from MediaPipe's face rotation	The old number captured the photo at about a 20° turn; people expected a full turn.

9. Open items and known gaps

Found while documenting the system, and listed openly so the team and the panel see the same picture.

Gap	Why it matters	Suggested fix
Users can read their own AI scores through the database. The website and the AI service's reply never show scores, but the database lets a logged-in user read their own verification record: face distance, live-scan result, QR result, and for a duplicate, the other account's ID.	Someone technical could use the scores to tune repeated attempts, and could learn which other account shares their face.	Make the table readable by admins only, and give users a small database function that returns just the status, reason, and dates.
Stored photos are not deleted when an account is deleted. Deleting an account removes its database records (including the face numbers), but the image files stay in the private bucket.	Conflicts with the right to erasure under RA 10173.	Delete the files through the Storage API whenever an account is deleted, plus a scheduled cleanup of files with no record.
No retention period. ID photos are kept until the account is deleted.	Data minimization suggests keeping them only as long as needed.	Decide a period (for example, delete the ID photos some days after approval) and keep only the result.
A recorded video could pass the live scan. The scan stops printed photos and photos on a screen, but not a video of the person played into the camera.	Impersonation with a victim's video.	Random prompts (for example, turn right first, or look up), blink or depth checks, or a paid liveness service.
Fake IDs can pass the AI. It cannot check issuer signatures or databases (section 4).	An admin has to catch them.	Ask the PSA for the public key used to sign PhilSys QR codes; with it the AI could confirm PhilSys IDs automatically. Keep the official manual checks for the other IDs.
Accuracy numbers come from small samples (about 60 public photos, labeled by another model).	Results may differ with real lighting, old IDs, and cheap cameras.	Run a pilot with volunteers, record the results, and adjust the settings.

10. References

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